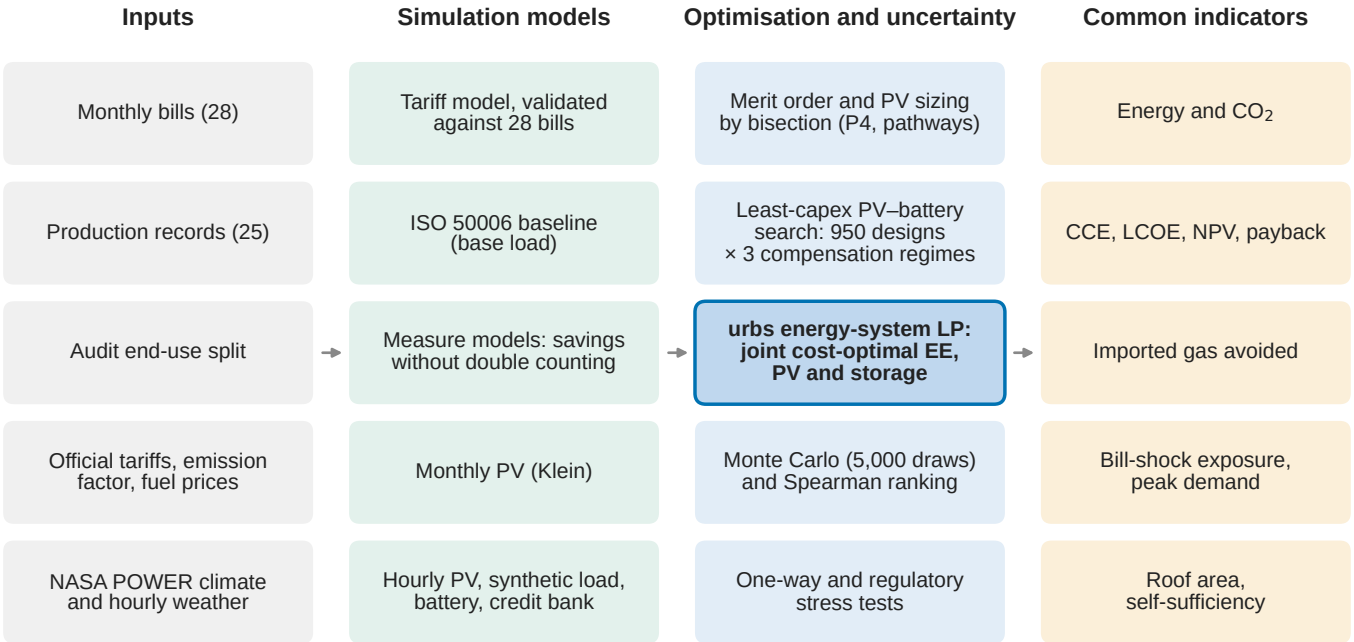
Inputs
Simulation models
Optimisation and uncertainty
Common indicators
Monthly bills (28)
Tariff model, validated against 28 bills
Merit order and PV sizing by bisection (P4, pathways)
Energy and CO_2
Production records (25)
ISO 50006 baseline (base load)
Least-capex PV–battery search: 950 designs × 3 compensation regimes
CCE, LCOE, NPV, payback
Audit end-use split
Measure models: savings without double counting
urbs energy-system LP: joint cost-optimal EE, PV and storage
Imported gas avoided
Official tariffs, emission factor, fuel prices
Monthly PV (Klein)
Monte Carlo (5,000 draws) and Spearman ranking
Bill-shock exposure, peak demand
NASA POWER climate and hourly weather
Hourly PV, synthetic load, battery, credit bank
One-way and regulatory stress tests
Roof area, self-sufficiency
EnergyPlus 25.1 verifies hourly PV yield and battery dispatch